

Features

General Functions:

- Full-duplex mode at 10/100/1000 Mbps and half-duplex mode at 10/100 Mbps
- IEEE 802.3-2002 compliant MII and GMII interfaces
- Independent 32-bit scatter-gather DMA with big/little endian operation
- Optional VLAN Q-Tag frame support
- CSMA/CD Protocol for half-duplex mode
- PAUSE frame based flow control in full-duplex mode
- MDIO/MDC management interface
- 802.3 compliant MIB, SNMP, RMON management support
- Configurable transmit and receive FIFOs
- Supports Jumbo frames
- Support magic packet and Wake-Up frames
- AHB master/slave interface

Transmit Functions:

- Variable length Inter Frame Gap (IFG) on back to back frame transmission
- Variable length (3, 5, 7 bytes) preamble generation
- Automatic generation of FCS and PAD
- Option to disable PAD or CRC32
- Option to stop frame retransmission on collisions

Receive Functions:

- Inter Frame Gap checking
- Preamble detection and stripping
- Flexible address filtering modes and inverse address filtering
- 64-bit hash table to filter multicast addresses
- Promiscuous mode of operation
- Reception of broadcast frames
- Automatic checking the FCS field, runt frames, and data field length
- Detection of MaxFrameLen frames, receive errors
- 32-bit status information on each receive frame

Flow Control Functions:

- Software controlled PAUSE control frame generation including multicast and unicast address.
- Automatic detection and checking of PAUSE frames
- Option to block PAUSE frames

Gigabit Ethernet - AHB IP Core

Overview

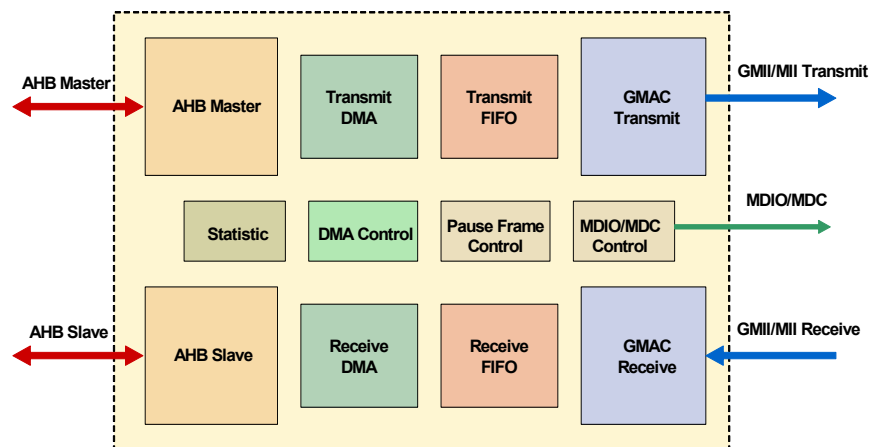
The Arasan Gigabit Ethernet Media Access Controller with AHB Interface IP core is compliant to the Ethernet/IEEE 802.3-2002 standard. The Gigabit Ethernet - AHB IP core provides a 10/100 Mbps Media Independent Interface (MII), a 1000 Mbps Gigabit Media Independent Interface (GMII), and an AHB master/slave interface. The IP core is designed for SoC and mobile applications such as integrated networking devices, PCI-Express Ethernet controllers, and Ethernet adapter cards.

The Gigabit Ethernet - AHB core supports half-duplex mode at 10/100 Mbps and full-duplex mode at 10/100/1000 Mbps. The IP core consists of two configurable FIFOs on both transmit and receive sides to handle the application's latency during frame transmission and reception. An AHB bus master and a 32-bit scatter-gather DMA transfer packets between the internal FIFOs and the host memory to enhance system performance.

The IP core supports 4-bit MII based 10/100 Mbps PHY and 8-bit GMII based 10/100/1000 Mbps PHY. A MDIO/MDC (Management Data Input/Output and Management Data Clock) management interface provides controlling and management functions to external PHY devices.

The Gigabit Ethernet - AHB core also provides enhanced programmable features for minimizing applications complexity and pre/post message processing. These features support MIB, SNMP, RMON, VLAN Q-Tag frame, and Jumbo frames. It also includes dynamic generation, checking, and stripping of FCS field, automatic pad field insertion, automatic retransmission and detection of collision frames, collision avoidance and handling. Other features are generation and decoding of PAUSE control frames, frame boundary delimitation, frame synchronization, and error detection.

Gigabit Ethernet - AHB IP Core IP Core Functional Block Diagram.



Gigabit Ethernet - AHB IP Core

GMII/MII Interface:

The GMII and MII interfaces are compliant to the IEEE 802.3-2002 clause 36 and clause 22 respectively. The interfaces can be connected to any standard compliant MII or GMII PHY. Full-duplex mode at 10/100/1000 Mbps and half-duplex mode at 10/100 Mbps are supported.

Gigabit Ethernet Receiver:

The receiver can be programmed to check minimum Inter Frame Gap (IFG) between frames and handles minimum IFG of 64 BT during back to back frame reception. It can also handle reception of frames with no preamble and only SFD. The receiver implements flexible address filtering mode with four 48-bit MAC addresses. A 64-bit hash table is included to filter multicast addresses.

Gigabit Ethernet Transmitter:

The transmitter supports variable length (64 BT, 96 BT, 128 BT, 256 BT) IFG on back to back frame transmission with default value of 128 bit times of IFG. It also supports variable length (3, 5, 7 bytes) preamble generation with default of 7 bytes of preamble. Other features include automatic generation of FCS and PAD during transmission, option to disable PAD and/or CRC32 insertion on transmission on a per frame basis, and option to shield the application for frame retransmission process on collisions during half-duplex mode.

The Gigabit Ethernet - AHB core IP Core is an RTL design in Verilog and VHDL that implements an Ethernet controller on an ASIC, or FPGA. The core includes RTL code, test scripts and a test environment for full simulation verifications.

MDIO/MDC Interface:

The MDIO/MDC interface is a 2-wire interface used to configure and control external IEEE802.3 compliant PHY devices using the MII Management Frame structure. The interface is defined in the IEEE802.3 Specification.

DMA:

The 32-bit scatter-gather DMA supports either the linked-list or ring (chained) descriptors. The skip length is programmable between descriptors when chained descriptors are used. Up to two buffers per descriptor are supported. The transmit DMA and receive DMA are operating independently. A programmable interrupt structure is used for the DMA. Big or little endian operation is supported.

AHB Interface:

The AHB master interface is used to transfer packets between the host memory and the internal FIFOs. The AHB slave interface is used to control the operation of the Ethernet controller and to program the registers inside the core.

Custom Interface:

The Gigabit Ethernet controller supports PowerPC, MIPS, SH3/SH4, 8051, and other processor buses besides the AHB. Custom buses with special requirements can also be implemented.

Benefits:

- Fully compliant core with proven silicon
- Premier direct support from Arasan IP core designers
- Easy-to-use industry standard test environment
- Unencrypted source code allows easy implementation
- Customer training available
- Reuse Methodology Manual guidelines (RMM) compliant verilog code ensured using Spyglass

Deliverables:

- RMM Compliant Synthesizable RTL design in Verilog
- Easy-to-use test environment
- Synthesis scripts
- Technical documents
- Sampled device drivers

Supported Platforms/Simulators:

- Platforms: Solaris, Unix, Linux and Win XP
- Verilog simulators: Synopsys VCS, Cadence NC-Verilog, MTI ModelSim-Verilog

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Data Sheet Links:

Gigabit Ethernet - AHB IP Core:
<http://www.arasan.com/datasheets/login.php>

For a complete directory of Arasan IPs, please visit: www.arasan.com

